



Yongzheng Zhang

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Lab: [Declarative Systems Lab](#)

EDUCATION

ShanghaiTech University

September 2024 - Present

Ph.D. Student in Computer Science and Technology

Shanghai, China

Research Interests: Explainable and Maintainable Network Configuration Verification, Synthesis, and Repair

Hainan University

September 2017 - June 2021

Bachelor of Engineering in Network Engineering

Haikou, China

RESEARCH INTERNSHIP

Tsinghua University

April 2024 - September 2024

Research Intern

Beijing, China (Remote)

Worked on **ClubHeap**, a high-speed, scalable programmable packet scheduler based on the PIFO (the push-in first-out queue) framework and priority queue structure with full pipeline architecture.

- Implemented the initial design of ClubHeap, a full pipeline architecture based on PIFO and priority queue.
- Developed a software simulator to verify the hardware design and conducted experiments in Vivado.

National University of Defense Technology

July 2021 - April 2022

Assistant Engineer

Changsha, China

Worked on **ResFuz**, a fuzzing engine for estimating worst-case resource usage by resource-usage-aware fuzzing.

- Responsible for managing heap and user-defined resources, and maintaining Shell and Coccinelle scripts.
- Engaged in scientific research and submitted artifacts for FASE'22, earning available and functional badges.

WORK EXPERIENCE

HUAWEI · Data Communication

February 2023 - May 2023

Software Testing Engineer

Nanjing, China

Worked on **NetEngine SD-WAN AR6700v/AR6700v-lite**, which supported resource pooling, on-demand subscription and other cloud-native technologies to build a secure and reliable multi-tenant operation solution.

- Responsible for automated testing of layer-2 and layer-3 network protocols (control-plane protocols).
- Developed automated testing scripts using the Pytest framework, and implemented general check methods.

PUBLICATIONS

Explainable Network Verification via Localized Subspecification

Yongzheng Zhang, Yaxuan Lin, Haoxian Chen, Ruize Ma, Amirmohammad Nazari, Mukund Raghothaman, and Peng Zhang.

In Proceedings of ACM SIGCOMM 2026.

WORKSHOP PAPERS AND POSTERS

Localized Explanations for Automatically Synthesized Network Configurations

Amirmohammad Nazari, Yongzheng Zhang, Mukund Raghothaman, and Haoxian Chen.

In Proceedings of ACM HotNets 2024.

Interpretable Network Verification via Subspecifications

Yongzheng Zhang, Haoxian Chen, Amirmohammad Nazari, and Mukund Raghothaman.
In Proceedings of ACM APNet 2025.

Characterizing Network Configuration Repair Spaces with Localized Subspecifications

Yongzheng Zhang and Haoxian Chen.
In Proceedings of ACM APNet 2026.

Explaining Network Configurations Under Failures via Localized Subspecifications

Yaxuan Lin, Yongzheng Zhang, and Haoxian Chen.
In Proceedings of ACM FMANO 2026.

TECHNICAL SKILLS

Programming Languages: Proficient in C/C++, familiar with Python, Java, P4, Shell, and Coccinelle.

Development Tools: Strong knowledge of the Linux environment, familiar with Vim, GCC, GDB, and Make.

AI-Assisted Software Development: Experienced in leveraging AI-assisted programming tools to improve development efficiency while maintaining code quality, scalability, testing, and long-term maintainability.

Analytical: Passionate about programming and research, skilled in identifying, analyzing, and solving problems.